

SENIOR ENGINEERING PORTFOLIO DELIVERABLE

PulseGrid

“The heartbeat of your campus, in one grid.”

Production-Grade Mini Student Information Management System (Mini-SIMS)

Problem Statement & Executive Vision

Addressing campus operational silos with unified instrumentation

OVERVIEW

The Operational Challenge

- ✦ **Fragmented Systems:** Academic attendance, fee ledgers, and timetables operate in isolated software silos.
- ✦ **Delayed Risk Detection:** At-risk students drop below threshold before intervention can occur.
- ✦ **Subpar Mobile Experience:** Legacy portals lack responsive mobile cockpit layouts for active faculty marking.

The PulseGrid Solution

- ✦ **Unified Role-Aware Cockpit:** Single dashboard tailored dynamically for Admins, Teachers, and Students.
- ✦ **Automated AI Risk Engine:** Instant weighted scoring analyzing attendance drops, fee delays, and lateness.
- ✦ **Mobile-First Instrumentation:** Fast, offline-capable UI with 44px tap targets and optimistic state sync.

System Architecture & Tech Stack

Modern production-ready full-stack web application

ARCHITECTURE

Frontend Core

- ✦ **Next.js 15 (App Router):** Server Components by default for optimal performance.
- ✦ **React 19 & TypeScript 5:** Strict type safety with `noUncheckedIndexedAccess`.
- ✦ **Tailwind CSS 4:** Modern design system with custom CSS variables.

State & Cache

- ✦ **Redux Toolkit:** Centralized slices for auth, attendance, finance, and schedules.
- ✦ **TanStack Query v5:** 30s background refetching with tab visibility pausing.
- ✦ **React Hook Form + Zod:** Strict runtime input schema validation.

Deployment & API

- ✦ **Vercel Serverless:** Built-in `/api/data/*` route handlers serving datasets.
- ✦ **Local Mock API:** JSON Server (`db.json`) running via `concurrently`.
- ✦ **PWA Shell:** Manifest, favicon ICO/SVG, offline page fallback.

Core Operational Modules

Instrument-grade user interfaces for everyday school operations

FEATURES

1. Attendance Register

Optimistic Present/Absent/Late toggle buttons with automatic rollback on network failure and immutable audit logging drawers.

2. Fees & Finance Portal

Ledger breakdown (paid/partial/overdue), payment modal, zero-balance state, and downloadable PDF receipts via `html2pdf.js`.

3. Conflict-Aware Timetable

Weekly class schedule grid with automated interval-overlap conflict detection math preventing double-booking.

4. Digital Noticeboard

Category-filtered announcement feed (Academic, Events, General) with role-gated admin composer.

AI-Powered Risk Scoring Engine

Proactive student intervention through weighted operational analytics

AI ANALYTICS

Risk Formula Specification

The operational risk engine calculates student vulnerability using a normalized weighted equation:

$$\begin{aligned} \text{Risk Score} = & (1 - \text{AttendanceRate}) * 0.5 \\ & + \text{OverdueFeeRatio} * 0.3 \\ & + \text{LatenessFactor} * 0.2 \end{aligned}$$

- ✦ **Low Risk Tier (Score < 0.25):** Normal academic standing.
- ✦ **Medium Risk Tier (0.25 - 0.40):** Flagged for advisor check-in.
- ✦ **High Risk Tier (Score > 0.40):** Urgent intervention watchlist.

Key Capabilities

- ✦ **Boundary Testing:** Fully validated against edge cases (100% attendance, zero fees).
- ✦ **Dual Fallback Execution:** Evaluates via serverless API (`/api/risk`) with client-side offline execution fallback.
- ✦ **Real-Time Watchlist:** Displays priority intervention alerts directly on the Admin & Teacher dashboard.

Design System & Accessibility (WCAG 2.1 AA)

High-contrast tokens, responsive layouts, and accessible UI controls

UI / UX

Curated Token Palette

Token Name	Hex Code	Usage
--pg-signal-blue	#1D4ED8	Primary actions & focus rings
--pg-pulse-teal	#0D7A6D	Present status & positive metrics
--pg-alert-amber	#9A5B00	Late status & warning alerts
--pg-risk-red	#B91C1C	Absent status & high risk flags

Accessibility Guarantees

- ✦ **Touch-First Controls:** 44×44px minimum tap targets across all interactive buttons.
- ✦ **Radix UI Backed:** Native keyboard navigation, focus traps, ARIA roles, and screen-reader labels.
- ✦ **Contrast Compliance:** Checked with `@axe-core/playwright` for 0 serious/critical violations.

Automated Verification & Testing Pipeline

100% test pass rate with strict quality gates

QUALITY ASSURANCE

100%

Line Coverage

Vitest suite covering pure logic math, risk calculations, and interval overlaps.

0

Lint Errors

TypeScript `tsc --noEmit` and ESLint checks passing with 0 warnings.

100%

E2E Pass Rate

Playwright + axe-core suite passing across Mobile & Desktop viewports.

PROJECT DELIVERABLE

PulseGrid is Live!

Explore the live production portal, view source code, or review complete documentation.

LIVE DEPLOYMENT URL

<https://pulse-grid-sigma.vercel.app>

GITHUB REPOSITORY

<https://github.com/Ebendttl/PulseGrid>

Thank you for reviewing PulseGrid Senior Engineering Capstone Project.